

Problem-Solving and Programming Completed SPARK Reference

SPARK Sports Academy - question-by-question completed guide

REFERENCE EXAMPLE ONLY

Use this document to study the wording, steps, structure and quality of completed work. Do not submit, rename or copy this SPARK project as your own SBA.

Project: SPARK Sports Academy

Sector: Sports

Questions being answered

Task A - Problem Definition and Algorithm

1. Write a clear problem statement for the following task: Calculate the amount each athlete should pay for a term and the total expected income for all athletes entered in one program run.

How SPARK completed it

- State what information the program receives.
- State the main calculation or decision the program performs.
- State the result the program must display.

Completed result

Problem statement: Calculate the amount each athlete should pay for a term and the total expected income for all athletes entered in one program run.

2. Identify the inputs, processes and outputs, then write pseudocode containing at least one selection and one repetition statement.

How SPARK completed it

- List inputs, processes and outputs before writing pseudocode.
- Use indentation to show the selection and loop structure.
- Keep variable names consistent from the algorithm through to the program.

Completed result

Inputs: Athlete name, Age, Programme code, Number of athletes

Processes: Select the monthly programme fee, Multiply monthly fee by 3, Apply a 10% junior discount when age is under 13, Add the final fee to a running total

Outputs: Athlete name, Programme code, Discount, Final term fee, Total expected income

Pseudocode:

```
START
SET grandTotal <- 0
INPUT athleteCount
FOR counter <- 1 TO athleteCount
INPUT athleteName, age, programmeCode
SET monthlyFee <- fee for programmeCode
SET termFee <- monthlyFee * 3
IF age < 13 THEN
SET discount <- termFee * 0.10
ELSE
SET discount <- 0
ENDIF
SET finalFee <- termFee - discount
SET grandTotal <- grandTotal + finalFee
DISPLAY athleteName, programmeCode, discount, finalFee
NEXT counter
DISPLAY grandTotal
STOP
```

Task B - Trace Table and Test Data

1. Create a trace table using at least three sets of test data. Include normal data, a boundary case and an invalid or unusual case where appropriate.

How SPARK completed it

- Choose values that exercise each branch of the selection statement.
- Follow the pseudocode in order and record important variable values.
- Calculate the expected output before running the program.

Completed result

The completed SPARK trace table includes the project's supplied normal examples and adds a boundary/validation discussion in the completed PDF.

Task C - Program Implementation

1. Implement the algorithm in the programming language selected by your centre. The SPARK Pascal reference is saved as AcademyFees_YourFullName.pas.

How SPARK completed it

- Create variables using suitable data types.
- Translate each pseudocode step into program statements in the same logical order.
- Compile or run often so errors are found in small sections.
- Use the same test data from the trace table.

Completed result

The SPARK reference implementation is AcademyFees_YourFullName.pas. The completed PDF includes the full source code.

2. Test the program and capture evidence that the expected and actual results agree.

How SPARK completed it

- Run normal test data first.
- Run the boundary case.
- Run an invalid or unusual input when the program includes validation.
- Capture screenshots showing both the entered data and the program result.

Completed result

The completed SPARK reference compares expected trace-table outputs with the program output and records the result of each test.

Task D - Documentation

1. Create ProgramDocumentation_YourFullName.pdf containing the problem definition, algorithm, source code, trace table, test data and screenshots of program execution.

How SPARK completed it

- Create a cover page and table of contents.

- Keep the section order the same as the task sequence.
- Use readable code formatting and captions for screenshots.
- Export the final document to PDF and check every page.

Completed result

Completed documentation filename: ProgramDocumentation_YourFullName.pdf. The SPARK section PDF demonstrates the required order and evidence.

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Problem definition

Calculate the amount each athlete should pay for a term and the total expected income for all athletes entered in one program run.

Input - Process - Output

Inputs	Processes	Outputs
Athlete name Age Programme code Number of athletes	Select the monthly programme fee Multiply monthly fee by 3 Apply a 10% junior discount when age is under 13 Add the final fee to a running total	Athlete name Programme code Discount Final term fee Total expected income

Pseudocode

```
START
SET grandTotal <- 0
INPUT athleteCount
FOR counter <- 1 TO athleteCount
    INPUT athleteName, age, programmeCode
    SET monthlyFee <- fee for programmeCode
    SET termFee <- monthlyFee * 3
    IF age < 13 THEN
        SET discount <- termFee * 0.10
    ELSE
        SET discount <- 0
    ENDIF
    SET finalFee <- termFee - discount
    SET grandTotal <- grandTotal + finalFee
    DISPLAY athleteName, programmeCode, discount, finalFee
NEXT counter
DISPLAY grandTotal
STOP
```

Trace table and test data

Name	Age	Code	Monthly Fee	Term Fee	Discount	Final Fee
Alicia Brown	12	SWM	9000	27000	2700	24300
Dario King	15	FTB	7500	22500	0	22500
Renee Blake	11	NET	7000	21000	2100	18900

Testing note

The normal data above should be supplemented with a boundary case and an invalid or unusual case where the program includes validation. Expected output should be calculated before the program is run.

Pascal reference implementation

```
program AcademyFees;
var
  i, athleteCount, age: integer;
  athleteName, programmeCode: string;
  monthlyFee, termFee, discount, finalFee, grandTotal: real;
begin
  grandTotal := 0;
  write('Number of athletes: '); readln(athleteCount);
  for i := 1 to athleteCount do
    begin
      write('Athlete name: '); readln(athleteName);
      write('Age: '); readln(age);
      write('Programme code: '); readln(programmeCode);
      if programmeCode = 'SWM' then monthlyFee := 9000
      else if programmeCode = 'FTB' then monthlyFee := 7500
      else if programmeCode = 'NET' then monthlyFee := 7000
      else monthlyFee := 8000;
      termFee := monthlyFee * 3;
      if age < 13 then discount := termFee * 0.10 else discount := 0;
      finalFee := termFee - discount;
      grandTotal := grandTotal + finalFee;
      writeln(athleteName, ' final fee: ', finalFee:0:2);
    end;
  writeln('Total expected income: ', grandTotal:0:2);
end.
```

Documentation order

- Cover sheet
- Table of contents
- Problem definition
- Algorithm
- Source code
- Trace table and test data
- Program execution screenshots
- Final review

Final section check

- Every question in the practice brief has a visible completed result.
- The output uses the same project facts as the other sections.
- The work has been tested or checked rather than assumed correct.
- This SPARK example is watermarked and is for study only. Build your own teacher-assigned SBA.

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